

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 33-2021: A 68-Year-Old Man with Painful Mouth Ulcers

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PRESENTATION OF CASE

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CME
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Dr. Ian D. Cooley (Medicine): A 68-year-old man was admitted to this hospital because of painful mouth ulcers and weight loss.

The patient had been well until 6 weeks before the current admission, when painful mouth ulcers and neck pain developed. He was evaluated at an urgent care center. He had reduced range of motion of the neck due to pain with flexion, extension, and rotation, but there was no spinal tenderness. There were small ulcers on the lower lip and gingivae. Cyclobenzaprine and acyclovir were prescribed.

Five weeks before the current admission, the patient was evaluated by his primary care physician in a clinic at this hospital. He reported that neck pain on the right side and oral ulcers had persisted and that several new oral ulcers had developed. He had no rash, eye pain, joint pain or swelling, dysuria or genital lesions, abdominal pain, or diarrhea. The examination was notable for several large ulcers on the lips and tongue (Fig. 1A). Treatment with acyclovir was continued. A direct immunofluorescence assay of a swab of an oral ulcer was negative for herpes simplex virus (HSV) types 1 and 2 and for varicella-zoster virus. Blood tests were negative for human immunodeficiency virus (HIV) types 1 and 2 and for syphilis; other laboratory test results are shown in Table 1. The patient was referred to the dermatology clinic for further evaluation of the oral ulcers.

Before the patient could be evaluated in the dermatology clinic, his mouth ulcers and pain worsened. Four weeks before the current admission, he presented to the emergency department of this hospital. Erythema and watering of the left eye had developed, as had new skin lesions on the trunk and upper face. The examination revealed multiple ulcers on the lips and tongue (Fig. 1B) and two pink crusted papules with hemorrhage on the right temple. There were scattered pustules on the lower back and right flank. An ophthalmologic examination revealed a pinguecula on the nasal side of the right eye with mild surrounding inflammation but was otherwise normal. Blood levels of electrolytes and glucose were normal, as were results of kidney- and liver-function tests. HLA typing was positive



Figure 1. Clinical Photographs.

Five weeks before admission, the patient had nonspecific aphthous-like ulceration and associated erythema of the ventrolateral tongue and labial mucosa (Panel A). Four weeks before admission, there was progression of the ulceration of the labial mucosa (Panel B). One week before admission, there was diffuse nonspecific ulceration and erythema of the ventrolateral tongue and labial mucosa (Panel C) and involvement of the soft palate (Panel D).

for the HLA-B51 allele; other laboratory test results are shown in Table 1.

The patient was evaluated by the rheumatology and dermatology services. The decision was made to pursue a skin biopsy before an oral biopsy. A biopsy of a pustule on the patient's back was performed, and examination of the specimen revealed mild spongiosis, neutrophilic granulomatous inflammation, and a superficial perivascular lymphohistiocytic infiltrate with neutrophils and occasional eosinophils. Colchicine was prescribed, along with oxycodone and lidocaine oral solution, for the treatment of a

working diagnosis of Behçet's disease. Outpatient follow-up with the rheumatology service was arranged.

Three days after discharge from the emergency department, the patient reported that the oral ulcers had not abated. Treatment with prednisone was started, and colchicine was continued.

Two weeks before the current admission, the patient was evaluated in the rheumatology clinic of this hospital. Prednisone treatment had led to only transient healing of the oral ulcers. The patient had pain with eating and speaking and had lost 10 kg of body weight in 3 weeks. The

Table 1. Laboratory Data.*

Variable	Reference Range, This Hospital†	5 Wk before Admission	4 Wk before Admission	2 Wk before Admission	On Admission
Hemoglobin (g/dl)	12.0–16.0	14.4	13.6	14.5	13.1
Hematocrit (%)	41.0–53.0	43.8	39.8	43.0	39.4
Mean corpuscular volume (fl)	80.0–100.0	95.2	94.8	92.5	93.4
White-cell count (per μ l)	4500–11,000	8150	8690	15,830	9510
Differential count (per μ l)					
Neutrophils	1800–7700	4560	4440	10,290	5250
Immature granulocytes	0–100	140	170	790	—
Lymphocytes	1000–4800	1410	1490	1380	1400
Eosinophils	0–900	1100	1200	690	1400
Basophils	0–300	70	70	190	—
Monocytes	200–1200	870	1320	2490	1470
Platelet count (per μ l)	150,000–450,000	255,000	235,000	301,000	172,000
C-reactive protein (mg/liter)	<8.0	10.6	21.5	—	—
Erythrocyte sedimentation rate (mm/hr)	0–20	13	16	—	—
Ferritin (μ g/liter)	20–300	—	338	—	—
Vitamin B ₁₂ (pg/ml)	>231	—	>2000	—	—
Folate (ng/ml)	>4.7	—	14.7	—	—
Cytomegalovirus					
DNA, PCR assay	Not detected	—	Not detected	—	—
IgG	—	—	Negative	—	—
IgM	—	—	Negative	—	—
Epstein–Barr virus					
DNA, PCR assay	Not detected	—	Not detected	—	—
IgG	—	—	Positive	—	—
IgM	—	—	Negative	—	—
Coxsackievirus antibody	<1:8	—	<1:8	—	—
Antinuclear antibody	—	—	Positive (1:320; homogeneous pattern)	—	—
Antineutrophil cytoplasmic antibody	Negative	—	Negative	—	—
Double-stranded DNA antibody	Negative at 1:10	—	Negative at 1:10	—	—
Anti-SSA (Ro) antibody	Negative	—	Negative	—	—
Anti-SSB (La) antibody	Negative	—	Negative	—	—
Anti-Smith antibody	Negative	—	Negative	—	—
Anti-U1-ribonucleoprotein antibody	Negative	—	Negative	—	—
Histoplasma antibody, serum	Negative	—	—	—	Negative
Lyme disease antibody	Negative	—	—	—	Negative
SARS-CoV-2, PCR assay	Negative	—	—	—	Negative
Bullous pemphigoid 180 IgG (RU/ml)	<20	—	—	—	57
Bullous pemphigoid 230 IgG (RU/ml)	<20	—	—	—	43
Desmoglein 1 antibody (RU/ml)	<20	—	—	—	<2
Desmoglein 3 antibody (RU/ml)	<20	—	—	—	15

* To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for folate to nanomoles per liter, multiply by 2.266. PCR denotes polymerase chain reaction, and SARS-CoV-2 severe acute respiratory syndrome coronavirus 2.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

neck pain, the eye watering, and several skin lesions had resolved. The examination revealed extensive ulceration of the upper and lower labial mucosa and of the ventrolateral surface of the tongue on both sides, as well as two focal ulcerations, measuring 0.5 cm in diameter, involving the soft palate on the left side (Fig. 1C and 1D). There were multiple purple-red, scaly papules involving the skin of the arms and lower legs. There was no evidence of a joint abnormality. Laboratory test results are shown in Table 1. Prednisone was switched to oral methylprednisolone, and treatment with amoxicillin–clavulanate was started for possible bowel-associated dermatosis–arthritis syndrome. Omeprazole, which the patient had taken for many years, was stopped because of peripheral eosinophilia. An evaluation in the oral medicine clinic at an affiliated hospital was arranged, with plans for oral and skin biopsies.

One week before the current admission, a biopsy of lower labial mucosa revealed squamous mucosa and parenchyma of the minor salivary glands, with mild nonspecific chronic inflammation. A skin biopsy was notable for epidermal ulceration with impetiginization and associated folliculitis. Dexamethasone oral rinses were prescribed, as was trimethoprim–sulfamethoxazole for prophylaxis against *Pneumocystis jirovecii* infection.

During the next 7 days, the ulcers worsened and the patient had ongoing pain that was unresponsive to oxycodone. He was unable to eat and was admitted to this hospital for pain control and further evaluation.

The patient's medical history included gastroesophageal reflux disease. Medications included methylprednisolone, colchicine, amoxicillin–clavulanate, trimethoprim–sulfamethoxazole, lorazepam, and oxycodone, as well as dexamethasone oral rinses. The patient was allergic to clarithromycin, which had caused a rash. He had quit smoking 30 years earlier, and he consumed 10 alcoholic drinks per week. There was no illicit drug use. The patient lived with his wife in a suburban area of New England. He had traveled to Western Europe, Mexico, and the Caribbean. He was employed part-time as a contractor. His family history included prostate cancer in his father and type 2 diabetes mellitus in his mother.

On examination, the temperature was 35.8°C,

the blood pressure 134/56 mm Hg, the heart rate 70 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 26.4. The patient looked uncomfortable, with an edematous tongue and multiple oral ulcers. There was no cervical lymphadenopathy or joint abnormality. Skin examination revealed a slightly raised pustule, measuring 2 mm in diameter, on the anterior aspect of the thigh. The remainder of the examination was normal. Laboratory test results are shown in Table 1.

Dr. Markus Y. Wu: Computed tomography (CT) of chest and abdomen, performed after the administration of intravenous contrast material, revealed subtle centrilobular ground-glass opacities in the upper lobes of the lungs, as well as prominent right hilar and left supraclavicular lymph nodes, each measuring 1 cm in diameter (Fig. 2).

Dr. Cooley: A diagnostic procedure was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Sook-Bin Woo: This 68-year-old man presented with a 6-week history of progressively worsening painful mouth ulcers associated with weight loss and the development of scaly papules on the skin. He had been treated with acyclovir, colchicine, prednisone, and amoxicillin–clavulanate, but the oral ulcers had not abated.

EVALUATION OF ORAL ULCERS

Evaluation of oral ulcers begins with determination of whether the patient has a single lesion or multiple lesions and whether the process is episodic or continuous (Fig. 3). This patient had multiple ulcers that had developed abruptly, with subsequent resolution of some ulcers and development of others at the same time, so that he was never ulcer-free. This combination of findings rules out conditions associated with a single episodic ulcer, including traumatic ulcers, which usually result from a defined traumatic event; solitary progressive ulcers due to cancer; and recurrent idiopathic aphthous ulcers, a common condition that almost always begins in the second or third decade of life and is characterized by ulcers that measure less than 1 cm in

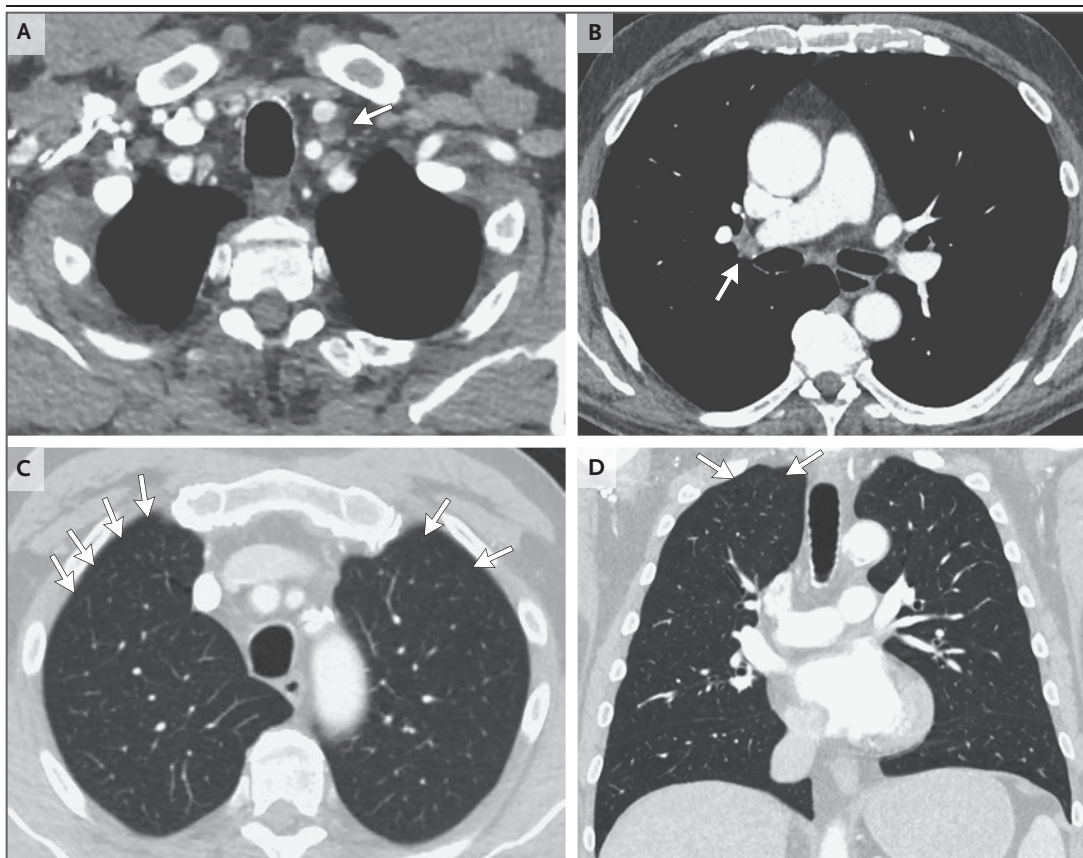


Figure 2. CT Scan of the Chest.

Images from a CT scan of the chest, obtained after the administration of intravenous contrast material, show a prominent left supraclavicular lymph node (Panel A, arrow) and a prominent right hilar lymph node (Panel B, arrow), each measuring 1 cm in short-axis diameter. In addition, there are subtle centrilobular ground-glass opacities (Panels C and D, arrows) predominantly involving the upper lobes.

diameter. Patients with recurrent idiopathic aphthous ulcers have ulcer-free periods between the episodes. There is an uncommon variant of the condition, known as severe recurrent aphthous ulcers or complex aphthosis, in which patients have continuous ulcers.¹ However, I am cautious about making a new diagnosis of recurrent idiopathic aphthous ulcers in patients older than 50 years of age, because the symptoms in question are more likely to represent another condition, such as aphthous-like ulcers that are caused by medications or are related to systemic disease.

SYSTEMIC DISEASE

The presence of oral ulcers is one of the defining criteria of Behçet's disease, but the oral ulcers associated with this disease tend to wax and wane and often respond to colchicine and gluco-

corticoid therapy.² In addition, this patient did not have genital, ocular, or typical cutaneous symptoms of Behçet's disease. Although HLA typing was positive for the HLA-B51 allele, this test is not highly specific for Behçet's disease and is positive in approximately 18% of the general population.³ Crohn's disease and infection with HIV type 1 are frequently associated with oral ulcers, but testing for HIV was negative and the patient had no symptoms suggestive of Crohn's disease.^{4,5}

VIRAL INFECTION

Could a viral infection be the cause of this patient's oral ulcers? The acute onset of multiple ulcers often occurs in patients with viral illnesses. Infection with HSV must be considered in this patient. Primary HSV infection is charac-

terized by a prodrome of fever and malaise, and painful clustered ulcers (ruptured vesicles) develop on both the nonkeratinized and the keratinized tissue of the tongue dorsum, hard palatal mucosa, and gingiva.⁶ These lesions often coalesce to form irregularly shaped ulcers that resolve within several weeks. This patient did not have a prodrome and the ulcers did not resolve over a 6-week period, so primary HSV infection is unlikely.

When reactivation of latent HSV infection occurs in healthy persons, clustered ulcers develop on the vermillion of the lip (herpes labialis) and involve only keratinized oral tissue. When HSV reactivation occurs in immunocompromised persons, ulcers may develop on any part of the oral mucosa, and multiple ulcers (similar to those seen in primary HSV infection) or a single ulcer may form.⁷ Although we do not know whether this patient was immunocompromised, testing of a specimen obtained from the ulcer was negative for HSV and varicella-zoster virus, and his condition did not improve with the use of acyclovir. Overall, HSV infection is an unlikely diagnosis in this case.

Mucocutaneous ulcers caused by other members of the herpesviridae family, such as cytomegalovirus or Epstein-Barr virus, are generally solitary and often occur in immunocompromised or immunosenescent persons; ulcers caused by deep fungal infections or syphilis may be multifocal.⁸⁻¹⁰ This patient had negative tests for cytomegalovirus IgG, treponemal antibodies, and histoplasma antibodies. On testing for Epstein-Barr virus, he was seropositive for IgG but negative for IgM, which is an antibody pattern consistent with previous infection.

AUTOIMMUNE VESICULOBULLOUS DISEASE

Could this patient's ulcers be ruptured blisters? The continuous occurrence of multiple oral ulcers, without ulcer-free periods, is a frequent finding in patients with autoimmune blistering diseases, such as pemphigus vulgaris. Because the mouth is a trauma-intense environment, vesicles and bullae typically rupture to form erosions and ulcers, and intact blisters are a rare oral finding.¹¹ Oral ulcers caused by pemphigus vulgaris can occur at any site but have a predilection for the palatal mucosa, and they may be several centimeters in diameter; these features are similar to those seen in this patient.¹² Many

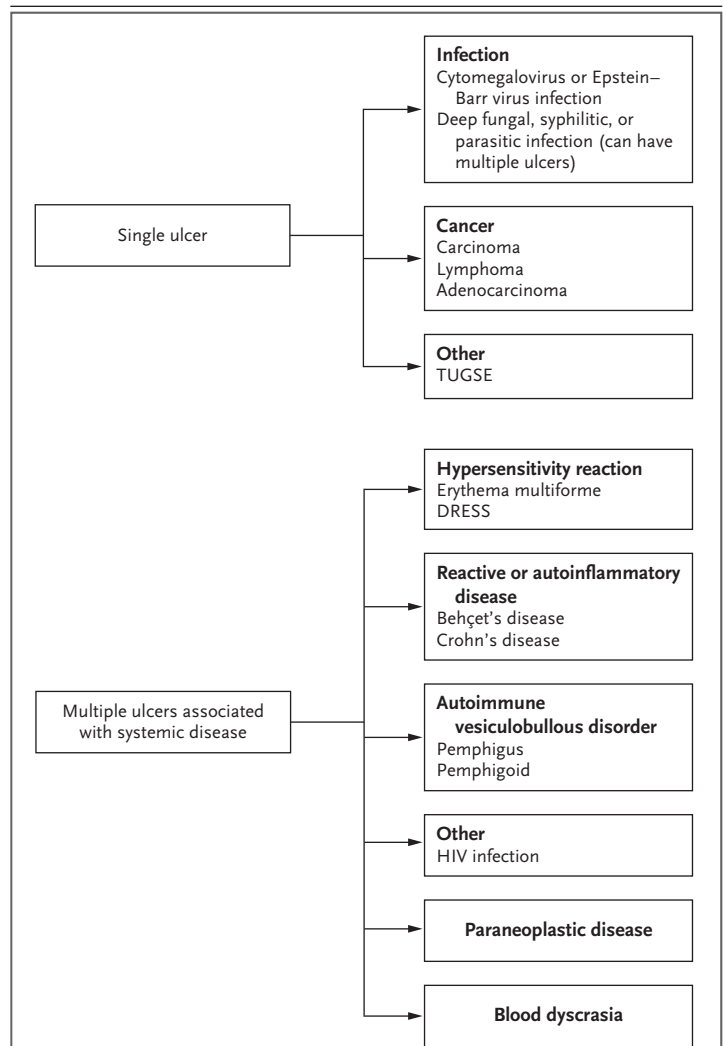


Figure 3. Evaluation of Chronic Oral Ulcers.

Single ulcers that persist for weeks or months are typically due to an infection or cancer; traumatic ulcerative granuloma with stromal eosinophilia (TUGSE) can mimic these diagnoses in severe cases. Multiple ulcers associated with signs of systemic disease can be due to hypersensitivity reactions, autoinflammatory diseases, autoimmune vesiculobullous disorders, human immunodeficiency virus (HIV) infection, paraneoplastic disease, or blood dyscrasia. DRESS denotes drug reaction with eosinophilia and systemic symptoms.

patients with pemphigus vulgaris initially present with oral ulcers, which can last for months before skin lesions develop.

Mucous membrane pemphigoid occurs in the oral cavity at a much greater frequency than pemphigus vulgaris. However, mucous membrane pemphigoid is more likely to cause desquamative gingivitis, characterized by diffuse gingival erosions and erythema, which was not

present in this patient.¹² This patient's ulcers could be a manifestation of oral bullous pemphigoid, but the skin findings are not consistent with this diagnosis.¹² Furthermore, the oral biopsy did not reveal either acantholysis (a finding characteristic of pemphigus vulgaris) or subepithelial clefting (a finding associated with pemphigoid). Unfortunately, the biopsy specimen lacked an intact epithelium–connective tissue interface and was suboptimal for evaluation. Although an additional biopsy could be considered, the absence of antibodies targeting desmoglein 1 and 3 rules out pemphigus vulgaris. The patient had low titers of antibodies targeting bullous pemphigoid antigens 180 and 230, but his overall clinical presentation and lack of response to glucocorticoids rule out pemphigoid.

PARANEOPLASTIC DISEASE

Patients with paraneoplastic pemphigus, also known as paraneoplastic autoimmune multi-organ syndrome, usually have extensive skin involvement that mimics pemphigus vulgaris, pemphigoid, graft-versus-host disease, erythema multiforme, and lichen planus. The cancers most frequently associated with paraneoplastic pemphigus are non-Hodgkin's lymphoma, chronic lymphocytic leukemia, B-cell lymphoma, and thymoma. Patients with paraneoplastic pemphigus may have extensive ulceration of the mouth, often with hemorrhagic crusts on the lips.¹³ This patient's oral ulcers could be consistent with oral involvement by paraneoplastic pemphigus, especially the pemphigus variant. However, he did not have a known underlying cancer and his skin lesions were fairly mild, which would be unusual for this diagnosis. If there is a strong suspicion for paraneoplastic pemphigus, an enzyme-linked immunosorbent assay or immunoprecipitation for envoplakin and periplakin are helpful tests, along with further evaluation for an occult cancer.

DRUG REACTION WITH EOSINOPHILIA AND SYSTEMIC SYMPTOMS

This patient's peripheral eosinophilia suggests the possibility of drug reaction with eosinophilia and systemic symptoms (DRESS),¹⁴ which occurs as a hypersensitivity reaction to medications such as allopurinol and anticonvulsants and less often to viral infections. Oral lesions are not frequently reported with DRESS. More-

over, liver injury is a hallmark of this disease, and this patient had normal liver function.

The finding of eosinophilia in this patient is worrisome, especially because he was receiving systemic glucocorticoid therapy. In addition, his ulcers did not abate despite methylprednisolone therapy, and he was noted to have lost 10 kg of body weight over a 3-week period. Taken together, these signs and symptoms are suggestive of a systemic disease such as a blood dyscrasia.

LEUKEMIA

Was this patient's presentation caused by a hematopoietic stem-cell disorder? The most common oral manifestation of acute leukemia is boggy, erythematous gingival hyperplasia that results from neoplastic infiltration of the gingiva. This finding is usually seen in patients with acute myeloid leukemia.¹⁵ However, patients with other myeloid neoplasms, especially those with eosinophilic leukemia, may also have oral ulcers that result from leukemic infiltration, which is likely to be associated with neutropenia.¹⁶

Although this patient did not have evidence of leukemic cells on oral biopsy, the diagnosis of leukemia would account for his weight loss, worsening ulcers that did not respond to therapy, and increasing eosinophilia despite treatment with glucocorticoids. To establish the diagnosis in this case, I would recommend a bone marrow biopsy.

DR. SOOK-BIN WOO'S DIAGNOSIS

Leukemia, perhaps of eosinophilic lineage.

DISCUSSION OF INITIAL MANAGEMENT

Dr. Nathaniel S. Treister: At the time of the evaluation in the oral medicine clinic, the patient's oral ulcerations had substantially worsened and were associated with intense pain. His mood was profoundly affected. Although there were several working diagnoses, it was unclear whether an oral mucosal biopsy would provide a definitive diagnosis. Therefore, our immediate management goal was to provide effective pain control. Opioids were of minimal benefit. The use of dexamethasone rinses, which are often prescribed

for the management of oral mucosal immune-mediated inflammatory disorders, exacerbated his symptoms. However, he reported that the use of mouthwash containing the combination of diphenhydramine, lidocaine, and magnesium hydroxide was a “lifesaver.” His condition resembled that of a patient with cancer who has severe oral mucositis.

Our approach was multifaceted, with attention to symptom control, diet and nutrition, and psychosocial considerations. Pain and its effect on the ability to function can lead to anxiety and depression, loss of appetite, and anorexia; screening for and treating anxiety and depression can be helpful. However, when the effect on oral intake is profound, as was the case in this patient, the need for supplemental feeding must be assessed. We recommended admission to the hospital for pain control and consideration of supplemental feeding.

Dr. Daniela Kroshinsky: The patient was seen by the dermatology service during his hospitalization, 6 weeks after the oral ulcers had developed. We recommended treatment with topical lidocaine applied before meals, a mucoadhesive gel, and gabapentin. The patient tried several varieties of dietary supplemental drinks, but ultimately, we recommended milkshakes as a way to increase caloric intake without exacerbating pain. With these measures, he was able to increase his oral intake enough to meet his caloric needs without the use of a nasogastric or gastric tube.

The lack of improvement while the patient received a prolonged course of high-dose glucocorticoid therapy, the persistent eosinophilia, and the focal, discrete nature of his lesions with the absence of gingival involvement made an autoimmune blistering disorder unlikely. We opted to wean the patient off glucocorticoid therapy in case he had a secondary infection that was impeding his healing and because the treatment appeared to have no effect on the oral lesions.

Next, we increased the colchicine dose, because the neutrophilic inflammation seen on biopsy and the discrete nature of the lesions were suggestive of aphthosis, possibly as a paraneoplastic phenomenon. The patient noted marked alleviation of his lesions and pain within 2 days. While he was admitted to the hospital, in collaboration with colleagues from hematology, peripheral blood was sent for mutational

analysis and a heme fusion assay, and a bone marrow biopsy was performed.

PATHOLOGICAL DISCUSSION

Dr. Robert P. Hasserjian: The diagnostic procedure was bone marrow biopsy and aspiration, as well as review of the peripheral blood smear and concurrent genetic testing of the blood and bone marrow. The patient had leukocytosis with monocytosis and eosinophilia. Review of the peripheral blood smear revealed mature-appearing monocytes; circulating blasts were not seen. The eosinophils included some forms with cytoplasmic vacuoles (Fig. 4A).

The bone marrow aspirate showed normal maturation in all three lineages, with 5% eosinophils and 1% blasts (Fig. 4B). The bone marrow-biopsy specimen was hypercellular for the patient's age and showed a predominance of myeloid elements, including eosinophils (Fig. 4C). On tryptase immunostaining (Fig. 4D), there was an increased number of mast cells, many of which were abnormally elongated; the mast cells had aberrant expression of CD25. Flow cytometric studies of both blood and bone marrow did not reveal any abnormal T-cell or B-cell populations.

On the basis of these findings, the differential diagnosis included a reactive eosinophilia, as well as myeloid neoplasms associated with eosinophilia. Given the increased number of mast cells, systemic mastocytosis was a diagnostic consideration. However, with systemic mastocytosis, mast cells typically form dense aggregates and there would usually be an elevated serum tryptase level.¹⁸

Myeloid neoplasms associated with clonal eosinophilia include several genetically defined entities, which underscores the importance of genetic testing in the workup of unexplained eosinophilia. Such entities include chronic myeloid leukemia, *BCR-ABL1*-positive (with the diagnosis based on the detection of *BCR-ABL1* rearrangement), and neoplasms of myeloid or lymphoid lineage with rearrangement of *PDGFRA*, *PDGFRB*, *FGFR1*, or *PCM1-JAK2*. If these genetic translocations are absent and other clonal genetic aberrations are present on karyotyping or next-generation sequencing or there are increased blasts, a diagnosis known as chronic eosinophilic leukemia, not otherwise specified, is ren-

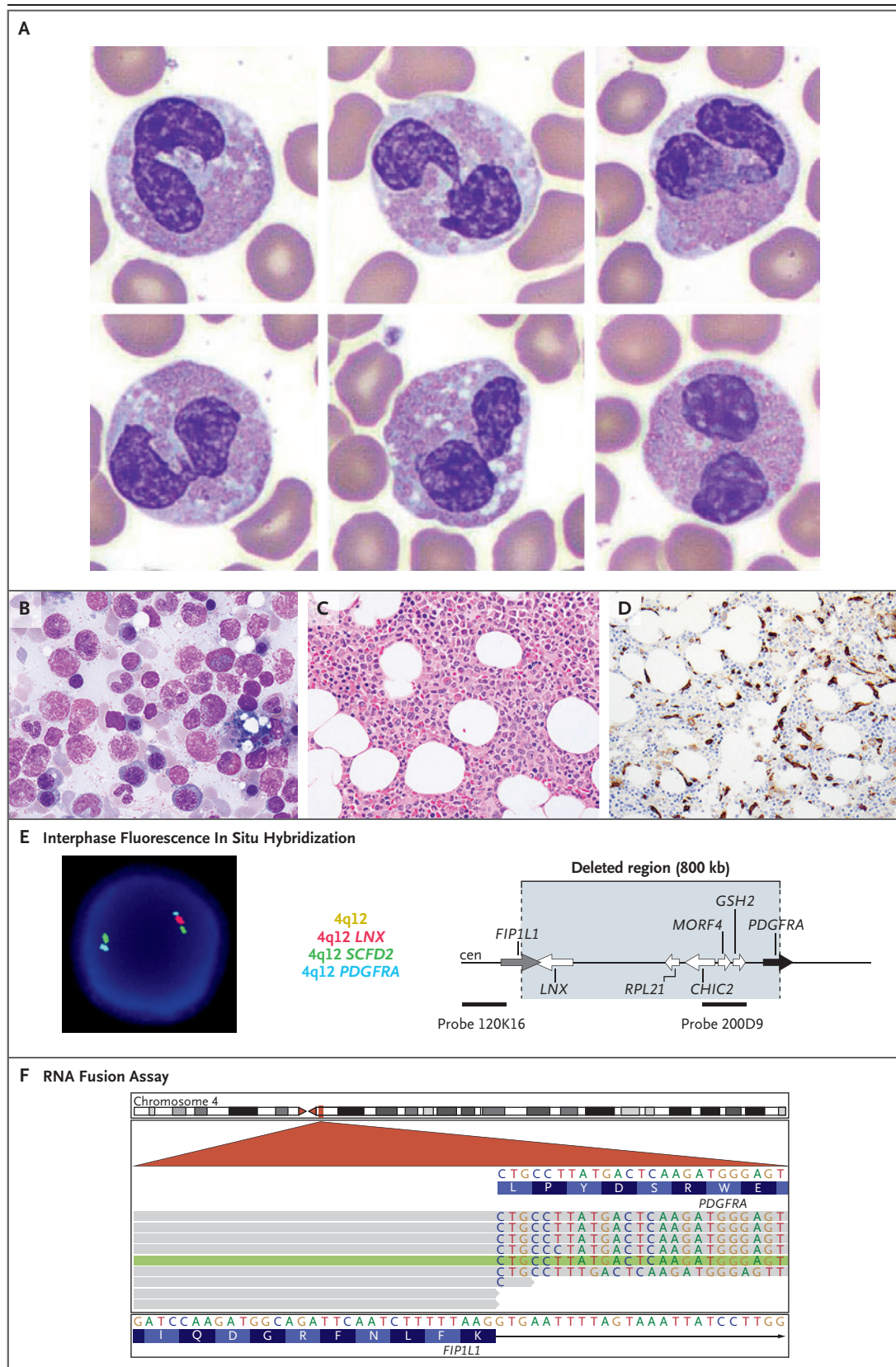


Figure 4 (facing page). Blood and Bone Marrow Specimens.

Wright–Giemsa staining of the blood (Panel A) shows an increased number of eosinophils, which have normal granulation and occasional cytoplasmic vacuolization. Wright–Giemsa staining of a smear of bone marrow aspirate (Panel B) shows normal maturation of myeloid and lymphoid lineages and a slightly increased number of eosinophils; there are no dysplastic features. Hematoxylin and eosin staining of a bone marrow–biopsy specimen (Panel C) shows hypercellularity for the patient's age (70% cells and 30% fat) and a relative predominance of myeloid cells, with a mildly increased number of eosinophils. Tryptase immunostaining of the bone marrow–biopsy specimen (Panel D) shows frequent interstitial mast cells, including abnormally elongated and spindled forms. On interphase fluorescence in situ hybridization performed on bone marrow (Panel E) with the use of probes to *LNK* (in red), *SCFD2* (in green), and *PDGFRA* (in aqua), the nucleus shows one chromosome with all three signals intact and another chromosome with intact *SCFD2* and *PDGFRA* signals but without the *LNK* signal. This finding indicates a deletion of the region between *FIP1L1* and *PDGFRA* on chromosome 4q12, as shown on the diagram on the right side of the panel, adapted from Cools et al.¹⁷; the abbreviation “cen” denotes the centromeric portion of the chromosome. On a next-generation sequencing–based RNA fusion assay performed on blood (Panel F), a transition from the sequence of exon 9 of *FIP1L1* (left side of the panel) to the sequence of exon 12 of *PDGFRA* (right side of the panel) indicates that a chimeric RNA transcript has resulted from the gene fusion.

dered. Unexplained persistent eosinophilia that is not associated with genetic proof of clonality or increased blasts but is associated with end-organ damage due to eosinophilia is placed in the category of idiopathic hypereosinophilic syndrome.

In this patient, the bone marrow karyotype was normal. However, the most common rearrangement partner of *PDGFRA* is *FIP1L1*, which is located in the same region on chromosome 4 as *PDGFRA*. These genes are juxtaposed through a small intrachromosomal deletion that cannot be detected on conventional karyotyping, and thus assessment for the *FIP1L1*–*PDGFRA* fusion must be performed with other genetic tests. An interphase fluorescence in situ hybridization study performed on a bone marrow specimen from this patient revealed a deletion of the *LNK* locus, which lies between *FIP1L1* and *PDGFRA* (Fig. 4E). In addition, a next-generation sequencing–based RNA fusion assay performed on a blood specimen revealed a chimeric transcript, which reflected an in-frame fusion between *FIP1L1* (exon 9)

and *PDGFRA* (exon 12) (Fig. 4F). A next-generation sequencing–based multiplex mutation screening assay (SNaPshot), performed to assess for mutations in 104 genes associated with myeloid and lymphoid neoplasms, detected the mutations *RUNX1* p.Phe158Tyr (c.473T→A) (variant allele frequency, 18%) and *RUNX1* p.Arg166HisfsTer11 (c.496_497insAC) (variant allele frequency, 11%). A *KIT* mutation was not detected, a finding that provided further evidence against the possibility of systemic mastocytosis. Overall, these findings confirm the diagnosis of a myeloid neoplasm with *PDGFRA* rearrangement, manifesting as chronic eosinophilic leukemia.

PATHOLOGICAL DIAGNOSIS

Myeloid neoplasm (chronic eosinophilic leukemia) with *PDGFRA* rearrangement.

DISCUSSION OF MANAGEMENT

Dr. Philip C. Amrein: Because hypereosinophilia can cause damage to many tissues, especially the heart, the patient underwent several studies, including electrocardiography, echocardiography, measurement of the troponin level, and pulmonary-function tests. He had no signs of cardiopulmonary damage. Treatment was commenced with imatinib, a tyrosine kinase inhibitor effective in patients who have neoplasms with *PDGFRA* rearrangement.

Within 1 month after treatment was initiated, the patient reported that he felt well; the mouth pain had resolved, he had regained some body weight, and the eosinophil count had normalized. The dose of imatinib was decreased. Two months after treatment was initiated, blood samples were sent for a repeat heme fusion assay and mutational analysis, and both the *FIP1L1*–*PDGFRA* fusion and the *RUNX1* mutations were undetectable.

At the most recent follow-up visit, after 6 months of treatment with imatinib, the patient continued to feel well and the dose of imatinib was further decreased. The low dose of imatinib has been continued indefinitely in this patient, because there have been only a few reports of patients remaining in remission after stopping imatinib treatment in this context, given that hypereosinophilia with the *FIP1L1*–*PDGFRA* fusion is quite rare.¹⁹

A Physician: Was it surprising that the patient's ulcers responded to a higher dose of colchicine, before treatment with imatinib?

Dr. Kroshinsky: After the dose of colchicine was increased, there was rapid and marked healing of the oral lesions. All the lesions resolved completely within 2 weeks, before the initiation of imatinib. The patient was successfully weaned off colchicine without a recurrence of oral lesions. I suspect that the oral lesions were due to an inflammatory process triggered by underly-

ing cancer, but the rapid response before the initiation of cancer therapy made paraneoplastic pemphigus unlikely.

FINAL DIAGNOSIS

Eosinophilic leukemia.

This case was presented at the Medical Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Lucas Massoth for assistance with Figure 4B.

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